Respiratory Rate Estimation from Bio-Signals

Group 14

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Abstract

Respiratory rate is a critical physiological parameter widely used in clinical monitoring and healthcare applications. In this project, respiratory rate estimation is performed from bio-signals using classical signal processing techniques. Photoplethysmography (PPG) signals were analyzed to extract respiratory information in a non-invasive manner.

A synthetic PPG signal was generated to simulate real-world conditions, including respiratory activity, cardiac interference, and sensor noise. The signal was preprocessed using baseline wander removal, bandpass filtering, and normalization to isolate the respiratory frequency band. Respiratory rate estimation was performed using both time-domain peak detection and frequency-domain analysis based on the Welch power spectral density method.

The results show that both approaches provide respiratory rate estimates close to the ground truth value, demonstrating the effectiveness of the implemented signal processing pipeline for biomedical applications.

1 Introduction

Respiratory rate is one of the fundamental vital signs used to assess human health. Deviations from normal respiratory patterns may indicate respiratory, cardiovascular, or neurological disorders. Conventional measurement techniques often rely on dedicated sensors, which may be inconvenient for continuous monitoring.

Recent advances in biomedical signal processing have shown that respiratory information can be extracted from photoplethysmography (PPG) signals. PPG is a non-invasive optical technique widely used in wearable devices, making it a suitable signal source for respiratory rate monitoring. This project focuses on estimating respiratory rate from PPG signals using time-domain and frequency-domain signal processing techniques.

2 Methodology

The proposed approach follows a structured signal processing pipeline. Initially, baseline wander was removed using a high-pass Butterworth filter to eliminate very low-frequency components caused by motion artifacts. Subsequently, a bandpass filter between 0.1 Hz and 0.5 Hz was applied to isolate the frequency range corresponding to normal human respiration (6–30 breaths per minute).

After preprocessing, respiratory rate estimation was performed using two complementary approaches. In the time domain, respiratory cycles were detected using peak detection, and the number of detected peaks was converted to breaths per minute. In the frequency domain, the Welch power spectral density (PSD) method was used to identify the dominant respiratory frequency. The estimated frequency was converted to BPM using:

$$RR_{BPM} = f_{peak} \times 60 \tag{1}$$

Using both approaches allows cross-validation of the estimated respiratory rate and improves reliability.

3 Data and Implementation

Due to the lack of real patient data, a synthetic PPG signal was generated to simulate realistic physiological conditions. The signal consists of a low-frequency respiratory component, a higher-frequency cardiac component, and additive Gaussian noise. The sampling frequency was set to 125 Hz, and the total signal duration was 60 seconds.

The project was implemented in Python using a modular structure. Separate modules were developed for data loading, signal preprocessing, respiratory rate estimation, and visualization. This modular design facilitated parallel development and clear division of responsibilities among group members.

4 Individual Contributions

The project was developed collaboratively, with each group member contributing to specific components:

- **Murat Emir Öncül**: Implemented the data loading module and generated the synthetic PPG signal used for testing.

- **Abdullah Tülek**: Developed the signal preprocessing pipeline, including baseline wander removal, bandpass filtering, and normalization.
- **Fatih Yılmaz**: Implemented the frequency-domain respiratory rate estimation using the Welch PSD method and authored the project report.
- **Ahmet Mirza Cangür**: Implemented the time-domain respiratory rate estimation using peak detection.
- **Melih Başar Ekmekci**: Implemented signal visualization and assisted with graphical interpretation and signal spacing.

5 Results and Discussion

The implemented system successfully estimated respiratory rate from the synthetic PPG signal using both time-domain and frequency-domain approaches. The estimated values were consistent with the ground truth respiratory rate used during signal generation.

The frequency-domain method demonstrated robustness against noise, while the time-domain approach provided intuitive interpretation based on detected respiratory cycles. The consistency between both methods confirms the reliability of the proposed signal processing pipeline.

6 Conclusion

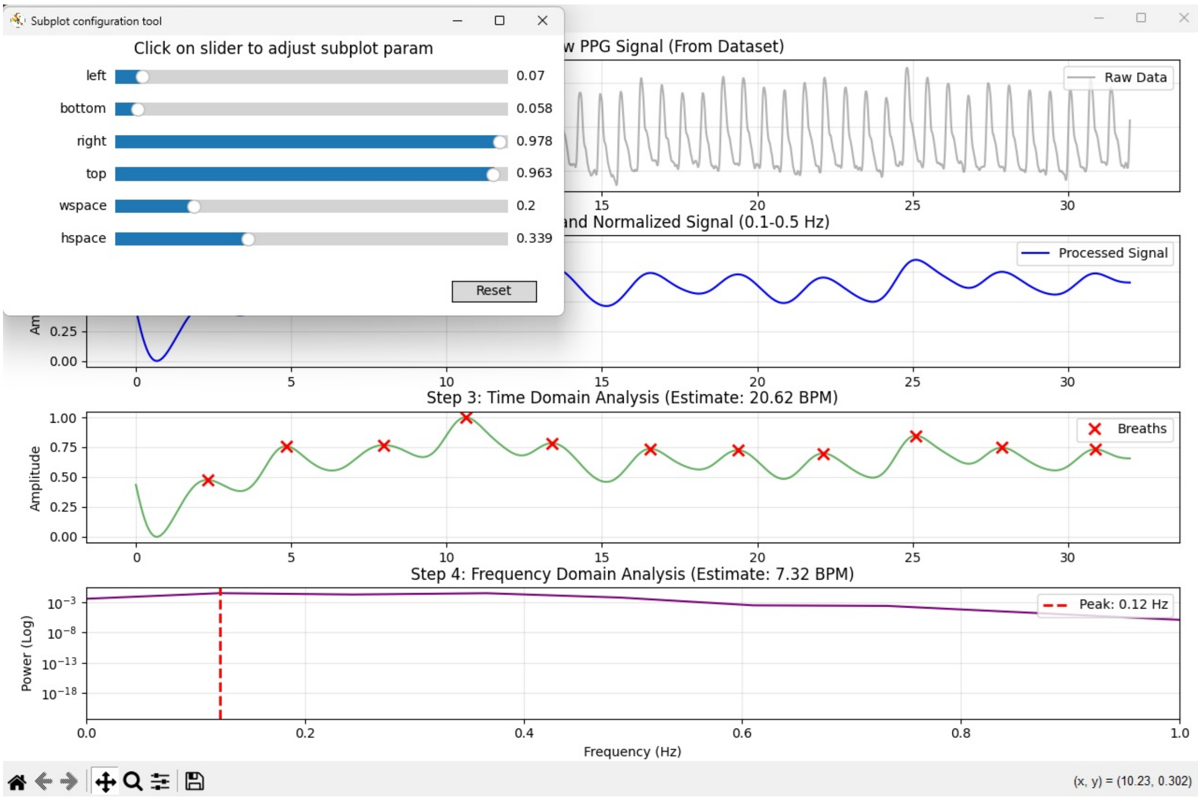
In this project, respiratory rate estimation from bio-signals was successfully implemented using digital signal processing techniques. Both time-domain peak detection and frequency-domain analysis were shown to be effective in estimating respiratory rate from PPG signals.

The project provides hands-on experience in biomedical signal processing and highlights the potential of non-invasive physiological monitoring. Future work may include applying the proposed methods to real datasets, improving robustness to motion artifacts, and exploring advanced data-driven approaches.

References

- SciPy Signal Processing Documentation
- NumPy Documentation
- Lecture Notes on Biomedical Signal Processing

7. Output Screenshots




```
--- Respiratory Rate Estimation Project ---
Dataset Loaded. Ground Truth RR: 21.375 BPM
  [INFO] Starting filtering process...
  [INFO] Signal cleaned and normalized with optimized parameters.
Time Domain Estimate: 20.62 BPM
Frequency Domain Estimate: 7.32 BPM
```